

Nakuja Project

Solid Propulsion Team

One-Grain Static Test — 0.1% Sodium Laureth Sulphate (SLS) Surfactant Trial

Document Version: 1.0 (Post-Execution Report)
Date of Test: Tuesday, 21st July, 2026
Motor Configuration: N4, 1-Grain, Aluminium Casing, BATES Geometry
Propellant Additive: 0.1% Sodium Laureth Sulphate (SLS) surfactant
Prepared by: Paschal Peters (Data Recorder / Timer)
Outcome: **FAIL** — Anomalous burn behaviour, no usable thrust data

1 Purpose

This report documents a static test conducted on a grain prepared with **0.1% Sodium Laureth Sulphate (SLS)** surfactant added to the propellant mix, as part of the team's ongoing investigation into surfactant-modified grain rheology (see the KNSB Grain Preparation Criteria, Revision A). While the surfactant addition produced a clear and immediate processing benefit — a significant improvement in castability — it also produced a severe, unintended alteration to the propellant's combustion behaviour. This report records the observed changes, the resulting test failure, and the implications for the permitted surfactant dosage going forward.

2 Test Overview

Motor	N4, 1-Grain, Aluminium Casing
Surfactant Additive	0.1% Sodium Laureth Sulphate (SLS)
Expected Burn Time (design target)	≈ 3 s
Observed Burn Time	Several minutes
Thrust Data Captured	None — load cell did not register a usable thrust curve
Outcome	Failure — combustion characteristics fundamentally altered

3 Observations

3.1 Processing Behaviour — Molten Mixture Consistency

The addition of 0.1% SLS produced a marked change in the rheology of the molten propellant mixture: rather than the expected thick, viscous consistency, the mixture became noticeably more **slurry-like**, with significantly reduced viscosity. This had one clear practical advantage — the mixture was considerably easier to mould into the desired grain shape, flowing more readily into the casting mold and conforming to the core geometry with less manual working.

3.2 Combustion Behaviour — Burn Rate and Duration

Despite the processing benefit, the grain's chemical/combustion behaviour was altered in the opposite direction to what was desired. Instead of the fast, powerful burn expected of the design (target burn time ≈ 3 s), the grain exhibited a **slow, consistent, gentle burn lasting several minutes** — an increase in burn duration best described as exponential relative to the target.

3.3 Instrumentation Impact — Load Cell / Thrust Curve

The burn was gentle enough that the **load cell failed to register any meaningful thrust signal**, and consequently no usable thrust curve was produced for this test. This is a direct consequence of the altered combustion regime: thrust output dropped low enough, for long enough, that it fell outside the range the load cell and logging setup were configured to resolve as a "static test" signal, rather than being a sensor fault in its own right.

4 Analysis: Why the Surfactant Dose Altered Combustion

The KNSB Grain Preparation Criteria (Revision A) specifies a **maximum permitted surfactant dose of 0.053 g per 100 g of dry mix** (0.053%), based on prior literature (EUCASS study) showing that a dose of 0.158 g/100 g caused self-extinguishment of the grain. The dose used in this test, **0.1% (0.1 g per 100 g)**, is roughly **double** the procedure's specified maximum, and sits within the range where the same literature reports severe combustion suppression.

This is consistent with what was observed: rather than acting purely as a processing aid (its intended role — reducing viscosity and improving castability, as documented in the grain preparation procedure), the surfactant at this concentration appears to have measurably interfered with the propellant's burn chemistry, suppressing burn rate to the point of an extended, low-intensity burn rather than the intended fast deflagration. The improved castability observed here is therefore not evidence that a higher surfactant dose is beneficial overall — it is a demonstration of the same trade-off the procedure already warns about: rheology improves while burn performance degrades, and the degradation becomes severe well before self-extinguishment.

5 Outcome Classification

This test is classified as a **FAIL**. Although the grain did ignite and combustion did proceed to completion without extinguishing outright, the burn profile bears no resemblance to the design intent: a multi-minute, low-thrust burn instead of a ~ 3 second high-thrust burn, with no thrust curve captured, does not constitute

a valid or usable static test result. The test does, however, provide a clear and valuable data point on the sensitivity of KNSB combustion behaviour to surfactant concentration.

6 Recommendations and Next Steps

- **Do not exceed the documented maximum surfactant dose of 0.053 g per 100 g of dry mix** in future grain batches. This test reinforces, rather than contradicts, that limit.
- Treat the 0.1% dose used here as a confirmed data point for the **Run B / Run D conditions** in the Grain Preparation verification test matrix (surfactant-only and surfactant+vacuum trials) — specifically, as an upper-bound warning case rather than a viable production dose.
- If further castability improvement is desired, investigate doses **below** the current 0.053% maximum in small increments, rather than above it, to find the point of diminishing returns before combustion is affected.
- Re-run the density and cross-section acceptance checks (per the Grain Preparation Criteria) on a retained sample of this grain, if available, to determine whether the altered burn was purely a surfactant/chemistry effect or was compounded by a porosity/density change from the altered rheology.
- For any future trial at or near this concentration, instrument the test with a lower-range or higher-sensitivity load cell setting (or a secondary slow-burn-capable logging configuration) so that anomalously long, low-thrust burns are still captured rather than lost entirely, as they carry diagnostic value even when the test itself is a failure.
- Update the Grain Preparation Criteria document's surfactant dose warning to reference this test as a second confirmed case of combustion degradation above the specified maximum, alongside the existing EUCASS reference.